

Probability and Statistics | Practice problems

Time Allowed: 120 minutes

Total Points: 100

Instructions:

- Answer all questions.
 - Show all work and justify your answers.
 - You may use a calculator.
 - Write your answers clearly.
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Question 1: Probability & Set Theory (10 points)

- Define the sample space for a single toss of a fair coin.
 - Let event A = “Head appears” and event B = “Tail appears.” Are A and B disjoint? Justify your answer.
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Question 2: Operations with Events (10 points)

Suppose events A and B satisfy:

$$P(A) = 0.4, \quad P(B) = 0.5, \quad \text{and} \quad P(A \cup B) = 0.7.$$

- Compute $P(A \cap B)$.
 - Compute $P(A^c \cap B^c)$.
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Question 3: Conditional Probability & Independence (10 points)

Two fair coins are tossed.

- List the sample space.
 - Let event C = “Exactly one head appears.” Compute $P(C)$.
 - Are the events “first toss is a head” and “exactly one head appears” independent? Explain your answer.
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Question 4: Discrete Random Variables (10 points)

Let X be a binomial random variable with parameters $n = 10$ and $p = 0.3$.

- a) Compute $P(X = 3)$.
 - b) Determine the expected value $E[X]$ and the variance $\text{Var}(X)$.
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Question 5: Continuous Random Variables (10 points)

Let Y be uniformly distributed over the interval $[2, 8]$.

- a) Write the probability density function (PDF) of Y .
 - b) Compute $P(3 \leq Y \leq 5)$.
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Question 6: Normal Distribution (10 points)

Suppose X is normally distributed with mean $\mu = 100$ and standard deviation $\sigma = 16$.

- a) Calculate the Z-scores for $X = 90$ and $X = 110$.
 - b) Determine $P(90 \leq X \leq 110)$ using the standard normal distribution.
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Question 7: Central Limit Theorem (10 points)

A random sample of 36 observations is drawn from a population with mean $\mu = 10$ and variance $\sigma^2 = 9$.

- a) Describe the sampling distribution of the sample mean $\bar{X}$.
 - b) Compute $P(\bar{X} \leq 11)$ using the Central Limit Theorem.
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Question 8: Confidence Intervals (10 points)

A random sample of 25 observations from a normally distributed population yields a sample mean $\bar{X} = 50$ and sample standard deviation $s = 4$.

Construct a 95% confidence interval for the population mean μ .

Question 9: Parameter vs. Statistic (10 points)

Explain the difference between a parameter and a statistic. Provide an example of each from a study involving the average income of a city's residents.

Question 10: Binomial Distribution & Normal Approximation (10 points)

A company claims that 70% of its products meet quality standards. In a quality control test, 100 products are randomly selected.

- a) Identify the appropriate probability distribution for the number of products meeting the standard.
- b) Calculate the probability that exactly 75 products meet the quality standard.
- c) Using a normal approximation (with continuity correction), find the approximate probability that at least 68 products meet the standard.